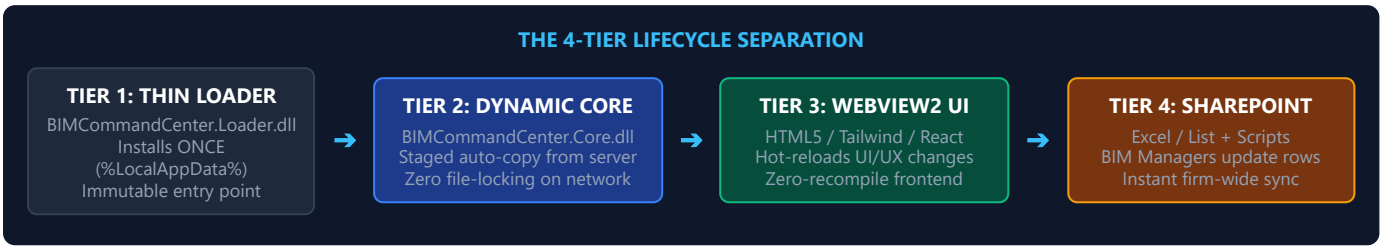
Enterprise Scalability & Rhino Portability

Architecture Validation, 4-Tier Separation & Cross-Platform Grasshopper Extension

SYSTEM SCALABILITY WHITEPAPER
Platforms: Autodesk Revit & McNeel Rhino 7/8
Status: Production Architecture | v1.0.0

1 Scalability Proof: The 4-Tier Decoupled Architecture

Traditional Revit plugins and Dynamo launchers fail to scale beyond a single project team because they are packaged as **monolithic DLLs**. Modifying a button icon, editing a script description, or releasing a new automation forces every workstation to close Revit and rerun an installer. In contrast, this solution enforces strict separation of concerns across four independent tiers:



2 Why This Architecture Scales to 500+ Seats

1. Zero Network File-Locking — STAGED LOADER

- **Problem:** Referencing DLLs directly from a network share (`\\server\BIM\`) locks files whenever *one* user opens Revit. IT cannot publish patches during work hours.
- **Solution:** The Loader queries the network share version, copies new releases to local staging in <2s, and loads locally. The server is never locked.

2. Dual-Runtime Compatibility — .NET 4.8 & .NET 8

- **Problem:** Revit 2025/2026 transitioned to .NET 8.0, breaking legacy .NET 4.8 add-ins. Firms running mixed-version projects encounter crashes.
- **Solution:** Multi-target build architecture produces separate native binaries for Revit 2023–2024 (net48) and 2025–2026 (net8.0-windows) from one unified codebase.

3. Zero-Privilege Installer — NON-ADMIN

- **Self-Service:** Installs into `%LocalAppData%` and `%AppData%\Autodesk\Revit\Addins` . Any architect or modeler can install in 5 seconds without IT elevation.
- **Mass Push:** Compatible with Microsoft Intune, SCCM, and GPO via silent flags (`/VERYSILENT /SUPPRESSMSGBOXES`).

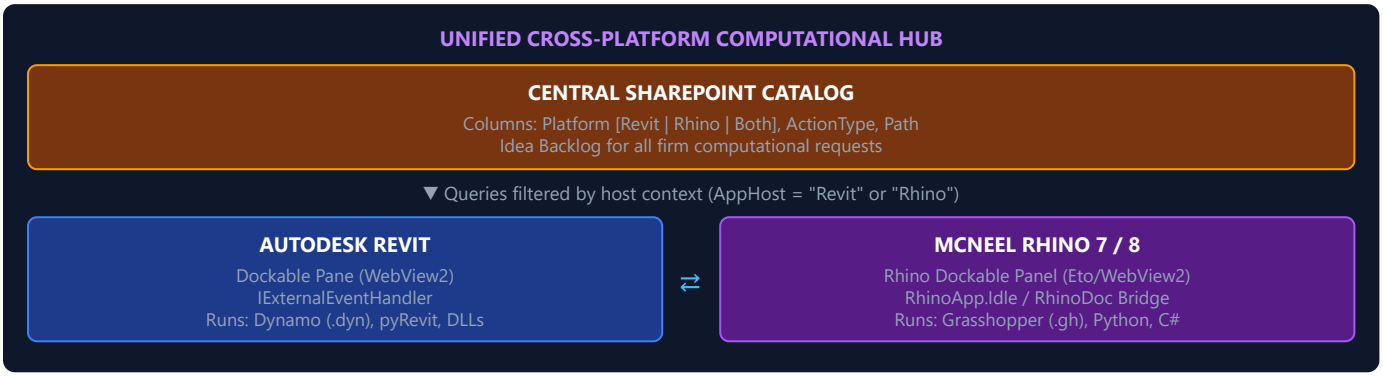
4. Offline Resilience & High Availability — CACHE-FIRST

- **Sub-10ms Startup:** Tool metadata is cached locally in `catalog_cache.json` . If SharePoint latency spikes or users are on site Wi-Fi, the UI launches instantly.
- **Vendored Assets:** Tailwind CSS and Lucide icons are bundled locally, eliminating external internet dependencies.

3 Portability: Extending the Architecture to Rhino & Grasshopper

Yes: 100% Compatible with Rhino 7 & Rhino 8 (Grasshopper Scripts)

Because the architecture completely decouples the **Web UI** and **SharePoint Catalog** from the CAD kernel, **the exact same system can power Rhino and Grasshopper**. You can even use a **single unified SharePoint catalog** across the entire firm: when opened in Revit it shows Revit/Dynamo tools; when opened in Rhino it shows Rhino/Grasshopper definitions.



4 Rhino & Grasshopper Implementation Blueprint

A. Rhino Dockable Panel (Host UI)

In Rhino, register a custom dockable panel using `Rhino.UI.Panels` hosting WebView2:

```
// RhinoCommon Plugin Entry Point
using Rhino.UI;

[Guid("9B2E3D4F-5A6C-7B8D-9E0F-1A2B3C4D5E6F")]
public class RhinoCommandCenterPanel :
  Eto.Forms.Panel, IPanelsMarker
{
  public RhinoCommandCenterPanel() {
    var web = new Microsoft.Web.WebView2.Wpf.WebView2();
    web.Source = new Uri("https://bimcc.local/index.html");
    // Wire up postMessage bridge
    Content = web.ToEto();
  }
}
```

B. Grasshopper Execution Engine (.gh)

Execute definitions headlessly or open on canvas via Grasshopper API:

```
// Headless Grasshopper Execution
using Grasshopper.Kernel;

public void RunGrasshopper(string ghPath, bool silent) {
  var docIO = new GH_DocumentIO();
  if (docIO.Open(ghPath)) {
    var doc = docIO.Document;
    if (silent) {
      // Recompute document silently
      doc.NewSolution(true);
    } else {
      // Show Grasshopper editor window
      Grasshopper.Plugin.GH_RhinoScriptInterface
        .ShowEditor();
    }
  }
}
```

5 Architecture Comparison: Revit vs. Rhino

Architectural Layer	Autodesk Revit Implementation	McNeel Rhino 7 / 8 Implementation
Host Shell / Plugin	`.addin` manifest + `IExternalApplication` DLL	`.rhp` plugin file or Yak package
Dockable Panel UI	Revit `IDockablePaneProvider` + WebView2	Rhino `Rhino.UI.Panels.RegisterPanel` + Eto / WebView2
Thread Safety Bridge	`IExternalEventHandler` via `ExternalEvent.Create()`	`RhinoApp.InvokeOnUiThread()` or `RhinoApp.Idle`
Computational Engine	Dynamo for Revit (`.dyn`) / pyRevit (`.py`)	Grasshopper (`.gh` / `.ghx`) / Rhino.Inside / Python 3
Headless Execution	DynamoRevit automation journal keys	`GH_Document.NewSolution(true)` in memory
Active View Capture	`Document.ExportImage()`	`RhinoView.CaptureToBitmap()`
Catalog & Ideas Backlog	Central SharePoint List / Excel (Graph API)	**Exact Same SharePoint List / Excel (Shared)**

6 Unified Firm-Wide Schema for Revit & Rhino

To support both Revit and Rhino tools from the same database, simply add an `AppHost` column to your SharePoint catalog:

Column	Type	Revit Usage	Rhino Usage
`AppHost`	Multi-Choice	Tagged: `Revit`	Tagged: `Rhino` (or `Both` for cross-platform workflows)
`ActionType`	Choice	`Dynamo`, `PyRevit`, `NativeAddin`	`Grasshopper`, `RhinoPython`, `RhinoCommand`
`TargetPath`	Text	`\Revit\RenumberDoors.dyn`	`\Rhino\FacadePanelization.gh`
`ExecutionMode`	Choice	`Headless` or `Player`	`Headless` (silent bake) or `Canvas` (open GH window)
`AppVersions`	Multi-Choice	`2023`, `2024`, `2025`, `2026`	`Rhino 7`, `Rhino 8`
`PackagesRequired`	Text	`Clockwork`, `Rhythm`	`Human`, `Elefront`, `Kangaroo2`, `Ladybug`

7 Enterprise Rollout & Governance Checklist

Phase 1: Revit Add-in Deployment (Ready Now)

- Distribute `BIMCommandCenter_Setup_v1.0.0.exe` to target users.
- Point `config.json` to your teammate's synced SharePoint catalog.
- Verify Dynamo Player launching and basic script execution.

Phase 2: Rhino & Grasshopper Integration

- Compile the Rhino plugin wrapper (`.rhp`) sharing the exact same frontend web app.
- Add Grasshopper script definitions to the central SharePoint folder.
- Distribute the Rhino plugin via standard company Yak package or file copy.

Strategic Takeaway: A Single Computational Nervous System

By implementing this architecture, your firm builds **one computational platform** rather than fragmented, isolated scripts. The same user interface, the same search bar, the same AI copilot, and the same crowdsourced idea backlog serve both your BIM documentation teams (in Revit) and your computational design teams (in Rhino & Grasshopper).